(i) Total Surface Area of the cube

= sum of area of the faces ($ABCD+EFGH+AEHD+BFGC+ABFE+CDHG$)

$=(a^2+a^2+a^2+a^2+a^2+a^2)$

$=6a^2$ sq. units

(ii) Lateral Surface Area of the cube

= sum of area of the faces ($AEHD+BFGC+ABFE+CDHG$)

$=(a^2+a^2+a^2+a^2)$

$=4a^2$ sq. units

If the side of a cube is a units, then,

(i) The Total Surface Area = $6a^2$ sq.units

(ii) The Lateral Surface Area = $4a^2$ sq.units

Thinking Corner

Can you get these formulae from the corresponding formula of Cuboid?

Example 7.6 Find the Total Surface Area and Lateral Surface Area of the cube, whose side is 5 *cm*.

Solution

The side of the cube (a) = 5 *cm*

Total Surface Area = $6a^2 = 6(5^2) = 150$ *sq. cm*

Lateral Surface Area = $4a^2 = 4(5^2) = 100$ *sq. cm*

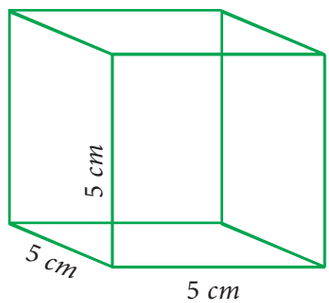


Fig. 7.15

Example 7.7 A cube has the Total Surface Area of 486 cm^2. Find its lateral surface area.

Solution

Here, Total Surface Area of the cube = 486 cm^2

$6a^2 = 486 \Rightarrow a^2 = \frac{486}{6}$ and so, $a^2 = 81$. This gives $a = 9$.

The side of the cube = 9 *cm*

Lateral Surface Area = $4a^2 = 4\times9^2 = 4\times81 = 324$ cm^2

7 Mensuration.indd 256 26-12-2019 14:16:33